

# Functional Data Analysis of Tecator Spectra Using FPCA and Depth-Based Outlier Detection

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June 24, 2026

## Abstract

This report applies Functional Data Analysis to the Tecator dataset, which contains near-infrared absorbance spectra of meat samples. Each sample is observed as an absorbance curve over wavelength, making the dataset naturally suitable for functional methods. The aim of the analysis is to represent the spectra as functional observations, study the main modes of variation by Functional Principal Component Analysis, and identify atypical spectral curves using depth measures and functional boxplots. The chemical variables fat, water and protein are then used to interpret the detected outliers.

The analysis shows that the spectra have a common functional structure but differ in level and shape. FPCA shows that most of the functional variability is explained by the first principal component, while later components provide secondary information about smaller modes of variation. Depth-based analysis identifies five outlying spectra. These outliers are not arbitrary: compared with the non-outlying samples, they have substantially higher average fat content and lower average water and protein content. This suggests that functional depth methods can detect meaningful spectral anomalies related to the composition of the samples.

## 1 Introduction

Functional Data Analysis is concerned with data in which each statistical observation can be represented as a function. Instead of treating an observation only as a finite-dimensional vector, FDA assumes that the observed values arise from an underlying curve  $x_i(t)$ , where  $t$  may represent time, wavelength, age, spatial position, or another continuous domain.

The Tecator dataset is a natural example of functional data. Each meat sample is described by a near-infrared absorbance spectrum observed over a range of wavelengths. Therefore, the statistical unit is not a single scalar value, but an entire curve. This motivates the

use of functional representation, smoothing, Functional Principal Component Analysis and depth-based methods.

Formally, the dataset can be considered as a sample of functions

$$X_1(t), X_2(t), \dots, X_n(t), \quad t \in \mathcal{T},$$

where  $n = 215$  is the number of meat samples and  $\mathcal{T}$  is the wavelength interval.

The main research questions of this report are:

1. Can Tecator absorbance spectra be suitably represented as functional observations?
2. What are the main modes of variation among the spectra according to FPCA?
3. Can depth measures and functional boxplots identify atypical spectral curves?
4. Are depth-based outliers related to fat, water, or protein content?

The report is structured as follows. Section 2 describes the Tecator dataset and explains why it is suitable for Functional Data Analysis. Section 3 presents the functional representation of the spectra using B-splines. Section 4 applies FPCA to study the main variation among curves. Section 5 introduces depth measures and functional boxplots for outlier detection. Section 6 interprets the detected outliers using the chemical variables. Section 7 discusses limitations and possible extensions. Section 8 concludes the report.

## 2 Dataset Description and Suitability for FDA

The Tecator dataset contains near-infrared absorbance spectra of meat samples. In this project, the dataset consists of 215 spectra, each observed at 100 wavelength points. For each sample, three scalar chemical variables are also available: fat, water and protein.

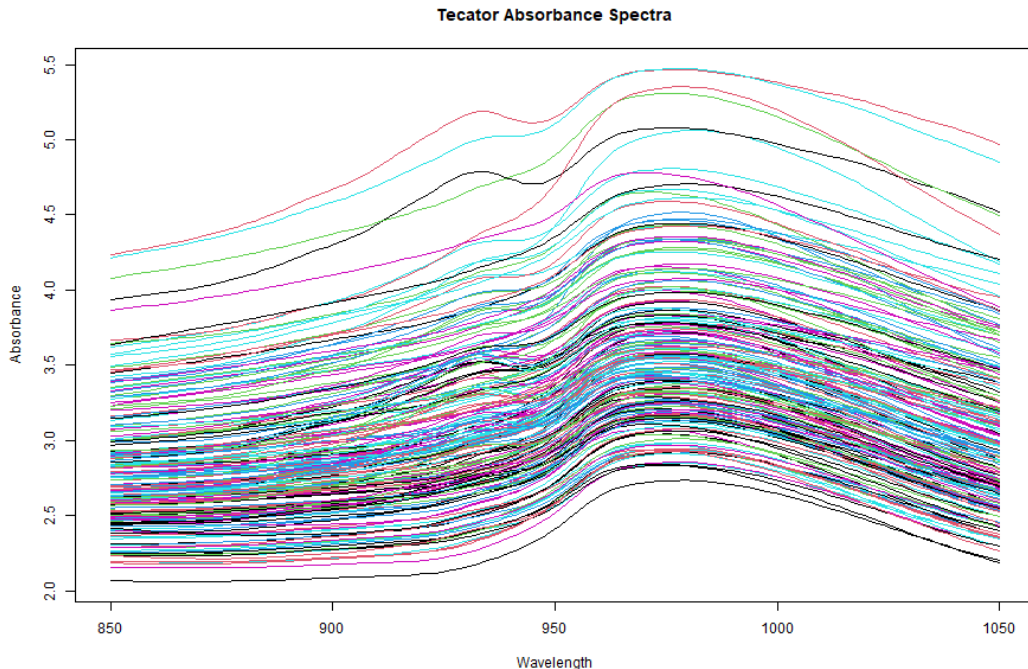
Each sample is represented by an absorbance curve measured over wavelength. Therefore, for sample  $i$ , the observation can be written as

$$x_i(t),$$

where  $t$  represents wavelength and  $x_i(t)$  represents absorbance at that wavelength.

This structure makes the dataset suitable for Functional Data Analysis. The object of interest is the whole spectral curve, not only isolated wavelength values. The shape, level and variation of the curves may contain information about the chemical composition of the samples.

The dataset is appropriate for this analysis for three reasons. First, the curves are observed on a common wavelength grid, which allows direct comparison between spectra. Second, the number of samples is large enough to apply exploratory FDA methods such as FPCA and depth ranking. Third, the chemical variables make it possible to interpret the detected functional outliers in terms of fat, water and protein content.



**Figure 1:** Raw Tecator absorbance spectra. Each curve represents one meat sample observed over wavelength.

The raw spectra show a common general pattern but also substantial variability in vertical level and shape. Some curves appear higher or lower than most others, suggesting that depth-based methods may be useful for identifying atypical spectra.

### 3 Functional Representation and Smoothing

The Tecator spectra are originally observed as discrete values at 100 wavelength points. However, Functional Data Analysis treats each sample as an underlying smooth function. Therefore, the first step is to transform the discrete observations into functional objects.

The observed spectrum can be represented as

$$y_i(t_j) = x_i(t_j) + \varepsilon_{ij},$$

where  $y_i(t_j)$  is the observed absorbance of sample  $i$  at wavelength  $t_j$ ,  $x_i(t)$  is the underlying smooth function, and  $\varepsilon_{ij}$  is an observational error term.

The representation used in this project is based on B-splines. Each smooth spectrum is approximated as a linear combination of basis functions:

$$x_i(t) \approx \sum_{k=1}^K c_{ik} \phi_k(t),$$

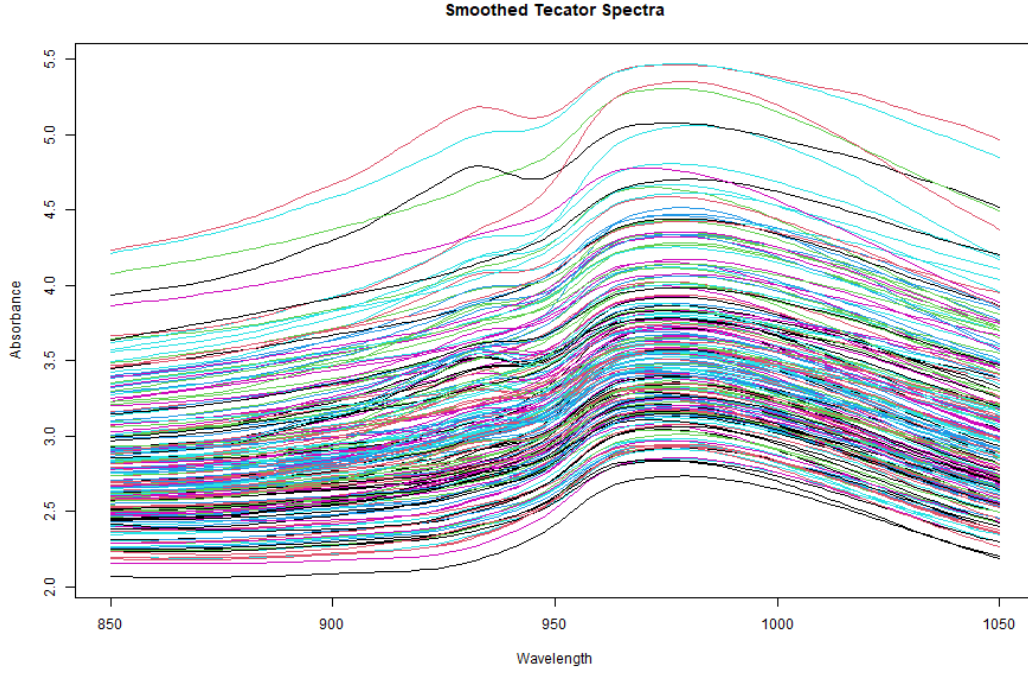
where  $x_i(t)$  is the absorbance function of sample  $i$ ,  $\phi_k(t)$  are B-spline basis functions,  $c_{ik}$  are the coefficients of sample  $i$ , and  $K$  is the number of basis functions.

In the implementation, a B-spline basis with 35 basis functions was used. The parameter `norder = 4` corresponds to cubic splines. Cubic splines provide a flexible but smooth representation of the spectra.

The choice of the number of basis functions controls the flexibility of the representation. A very small number of basis functions could oversmooth the spectra and remove relevant local features. A very large number could follow noise too closely. The choice of 35 basis functions provides a reasonable balance for this exploratory analysis.

Registration was not considered necessary in this project. Registration is important when curves are misaligned along the argument axis, for example when similar events occur at different times. In the Tecator dataset, all spectra are observed on the same wavelength grid. Therefore, the curves are already aligned by wavelength, and the main issue is functional representation rather than phase alignment.

All computations were performed in R using the packages `fda` and `fda.usc`.



**Figure 2:** Smoothed Tecator spectra obtained using a B-spline basis representation.

The smoothed spectra preserve the main shape of the raw curves while providing a functional representation suitable for FPCA and depth analysis. Since the original spectra are already relatively smooth, the B-spline representation mainly provides a functional form for the subsequent analysis rather than radically changing the observed data.

## 4 Functional Principal Component Analysis

Functional Principal Component Analysis is the functional analogue of classical PCA. Its goal is to identify the main directions of variation among curves. Instead of summarizing variability using eigenvectors in finite-dimensional space, FPCA summarizes functional variability using principal component functions, also called harmonics.

The sample mean function is defined as

$$\bar{x}(t) = \frac{1}{n} \sum_{i=1}^n x_i(t).$$

The covariance function describes how values of the curves vary jointly at two wavelengths:

$$v(s, t) = \frac{1}{n} \sum_{i=1}^n (x_i(s) - \bar{x}(s)) (x_i(t) - \bar{x}(t)).$$

The functional principal components are obtained as eigenfunctions of the covariance operator:

$$\int_{\mathcal{T}} v(s, t) \psi_j(s) ds = \lambda_j \psi_j(t),$$

where  $\lambda_j$  is the eigenvalue associated with the  $j$ -th principal component function  $\psi_j(t)$ .

A functional observation can then be approximated as:

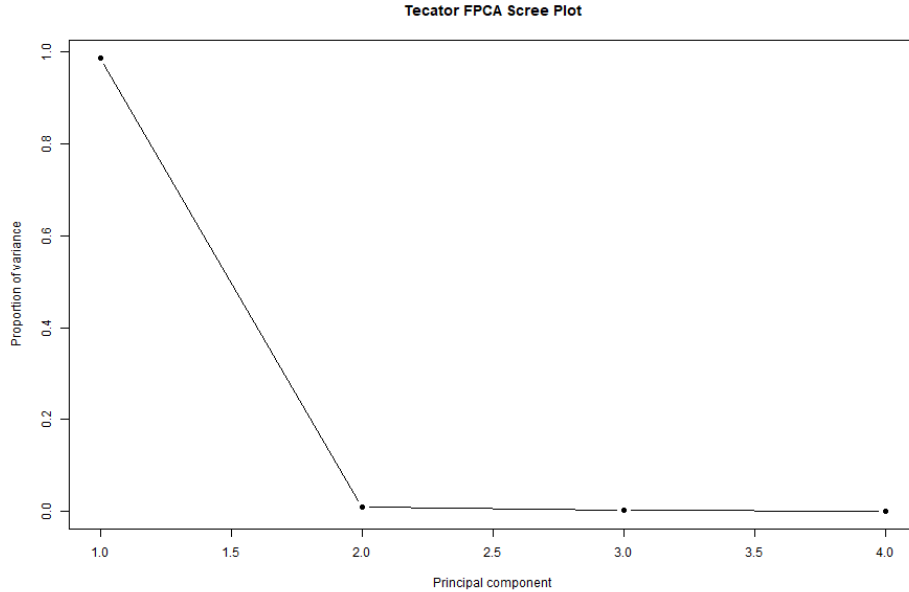
$$x_i(t) \approx \bar{x}(t) + \sum_{j=1}^L \xi_{ij} \psi_j(t),$$

where  $\bar{x}(t)$  is the mean function,  $\psi_j(t)$  is the  $j$ -th functional principal component,  $\xi_{ij}$  is the FPCA score of sample  $i$  on component  $j$ , and  $L$  is the number of retained components.

The FPCA score of curve  $i$  on component  $j$  is given by

$$\xi_{ij} = \int_{\mathcal{T}} (x_i(t) - \bar{x}(t)) \psi_j(t) dt.$$

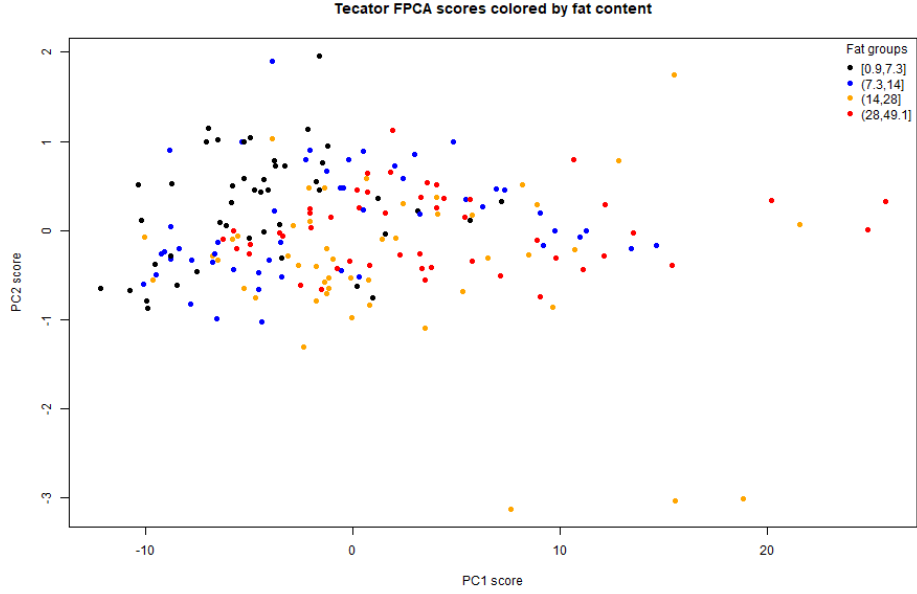
The FPCA scores summarize each spectrum in a low-dimensional way. This makes it possible to visualize the spectra in a two-dimensional score plot and to study how the main directions of variation are associated with chemical composition.



**Figure 3:** FPCA scree plot showing the proportion of variance explained by the first functional principal components.

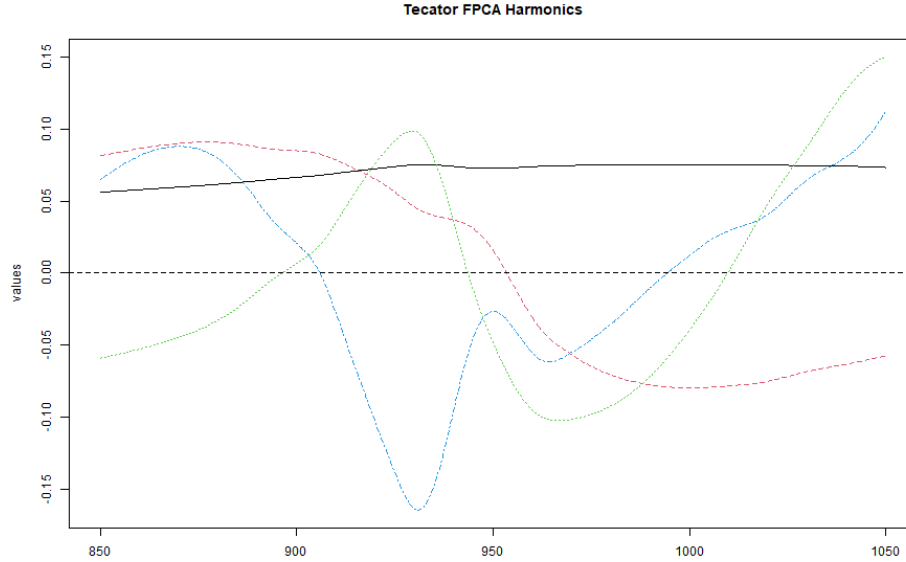
The scree plot shows the proportion of total functional variability explained by each component. In this analysis, the first principal component explains almost all of the functional

variation, while the following components explain much smaller proportions. This suggests that a large part of the variability among spectra is related to a dominant functional direction, most likely associated with overall level and global spectral shape.



**Figure 4:** FPCA score plot colored by fat content groups. Each point corresponds to one spectrum.

The score plot represents each spectrum using its first two FPCA scores. Coloring the points by fat content provides a visual indication of whether spectral variation is associated with chemical composition. The plot suggests a gradual relationship between spectral variation and fat groups, especially along the first component.



**Figure 5:** FPCA harmonics representing the main directions of functional variation around the mean function.

The harmonics represent the functional directions of variation. They show how spectra differ from the mean function across wavelengths.

The correlation between the first four FPCA scores and the chemical variables was also computed.

| Component | Fat   | Water | Protein |
|-----------|-------|-------|---------|
| PC1       | 0.44  | -0.48 | -0.26   |
| PC2       | -0.14 | 0.13  | 0.36    |
| PC3       | 0.62  | -0.65 | -0.47   |
| PC4       | -0.53 | 0.47  | 0.52    |

**Table 1:** Correlations between FPCA scores and chemical variables.

The correlation table suggests that some FPCA scores are associated with chemical composition. PC1, which explains almost all of the functional variation, is moderately positively associated with fat and negatively associated with water. Some later components, especially PC3 and PC4, show stronger correlations with the chemical variables. However, since these components explain a much smaller proportion of total functional variability, their interpretation should be treated as secondary and exploratory.

This does not imply a causal relationship. However, it indicates that FPCA captures meaningful spectral variation associated with the physical composition of the samples.

## 5 Depth Measures and Functional Boxplot

In ordinary univariate data, observations can be naturally ordered from smallest to largest. This makes it straightforward to define concepts such as median, quartiles and outliers. For functional data, this is more difficult because curves do not have a natural total ordering.

Depth measures address this problem by assigning each curve a centrality value. A curve with high depth is considered central with respect to the sample, while a curve with low depth is considered more outlying.

In this project, the analysis is based on Modified Band Depth. The general idea is that a curve is central if it often lies inside bands generated by other curves. For a sample of functions  $X_1, \dots, X_n$ , the band generated by two curves  $X_{j_1}$  and  $X_{j_2}$  is defined pointwise as

$$B_{j_1, j_2}(t) = [\min\{X_{j_1}(t), X_{j_2}(t)\}, \max\{X_{j_1}(t), X_{j_2}(t)\}].$$

The Modified Band Depth of a curve  $X_i$  can be written as

$$MBD(X_i) = \binom{n}{2}^{-1} \sum_{1 \leq j_1 < j_2 \leq n} \frac{1}{|\mathcal{T}|} \int_{\mathcal{T}} \mathbb{I}\{\min(X_{j_1}(t), X_{j_2}(t)) \leq X_i(t) \leq \max(X_{j_1}(t), X_{j_2}(t))\} dt.$$

Here  $\mathbb{I}\{\cdot\}$  is an indicator function. The value of  $MBD(X_i)$  is high when the curve lies inside many bands for a large part of the domain. In the implementation, the integral over  $\mathcal{T}$  is approximated using the observed wavelength grid.

Intuitively, a central curve lies inside many bands for many wavelengths, while an outlying curve lies outside many bands. The deepest curve can be interpreted as a functional median.

After ranking curves by depth, the functional boxplot is constructed. The deepest curve is the functional median. The central region is formed by the 50% deepest curves. Let  $C$  denote the set of the 50% deepest curves. The central envelope is defined pointwise by

$$L(t) = \min_{i \in C} X_i(t), \quad U(t) = \max_{i \in C} X_i(t).$$

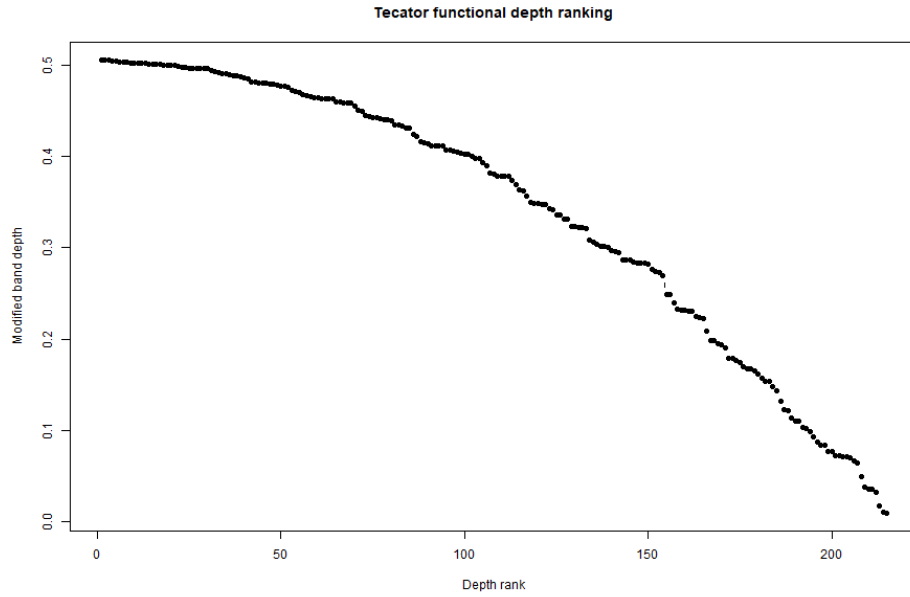
The functional boxplot fences are constructed as

$$L_f(t) = L(t) - 1.5(U(t) - L(t)),$$

$$U_f(t) = U(t) + 1.5(U(t) - L(t)).$$

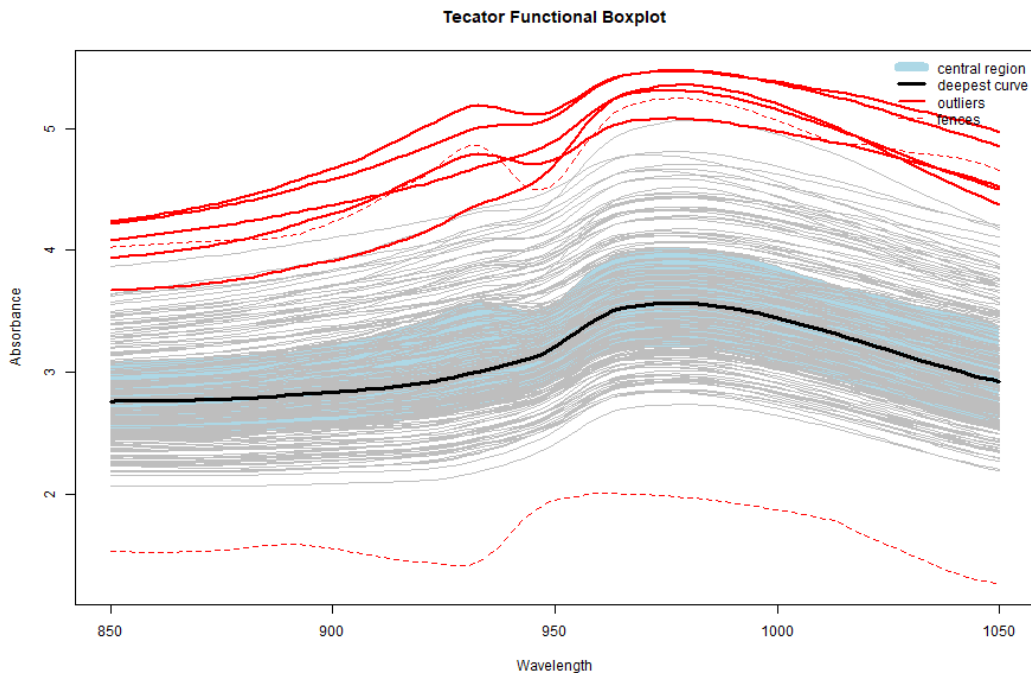
A curve is flagged as an outlier if it crosses these fences for at least one value of  $t$ . This is

analogous to the outlier rule in a classical boxplot, but adapted to curves.



**Figure 6:** Functional depth ranking of the Tecator spectra using Modified Band Depth. Curves are ordered from most central to least central.

The depth ranking shows the spectra ordered from most central to least central. Curves with small depth values are candidates for atypical behavior.



**Figure 7:** Functional boxplot of Tecator spectra. The central region is shown in blue, the deepest curve in black, the fences as dashed red lines, and detected outliers in red.

The functional boxplot identified five outlying spectra:

43, 44, 99, 140, 185.

These curves are not merely isolated numerical observations. They are entire spectra whose shape or level differs from the central functional pattern.

## 6 Chemical Interpretation of Depth-Based Outliers

A key part of the analysis is to determine whether the functional outliers correspond to meaningful chemical differences. The depth-based method detected five outlying samples. Their chemical values are shown in Table 2.

| Sample | Depth  | Fat  | Water | Protein |
|--------|--------|------|-------|---------|
| 43     | 0.0384 | 48.2 | 40.2  | 11.0    |
| 44     | 0.0103 | 48.5 | 39.5  | 12.0    |
| 99     | 0.0322 | 17.6 | 64.4  | 18.0    |
| 140    | 0.0502 | 22.4 | 62.4  | 11.5    |
| 185    | 0.0175 | 34.8 | 50.3  | 14.6    |

**Table 2:** Chemical composition of the depth-based outliers.

The average chemical composition of non-outlying and outlying samples is shown in Table 3.

| Group        | Fat   | Water | Protein |
|--------------|-------|-------|---------|
| Non-outliers | 17.76 | 63.49 | 17.78   |
| Outliers     | 34.30 | 51.36 | 13.42   |

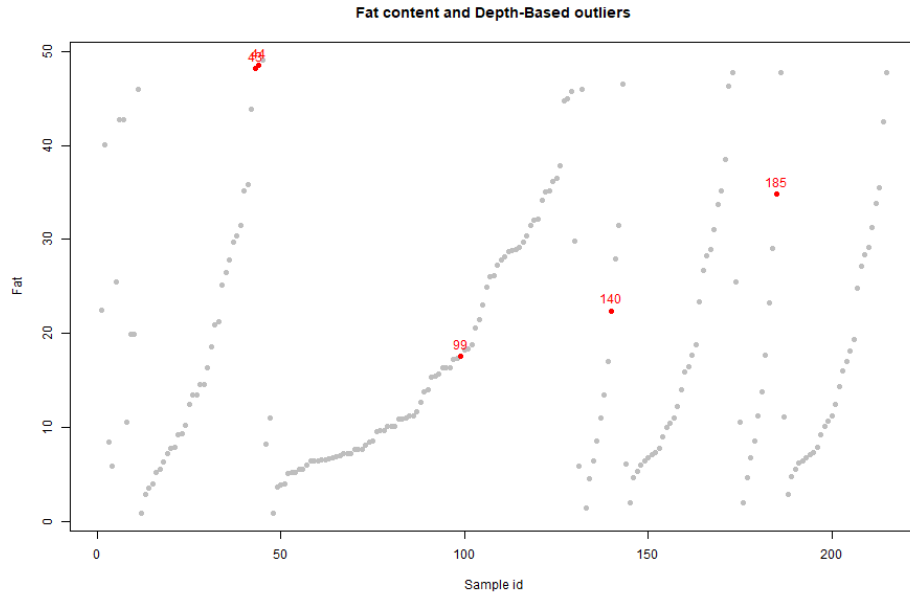
**Table 3:** Average chemical composition of non-outlying and outlying samples.

The detected outliers have much higher average fat content and lower average water and protein content than the non-outlying samples. This suggests that the functional outliers correspond to meaningful chemical differences.

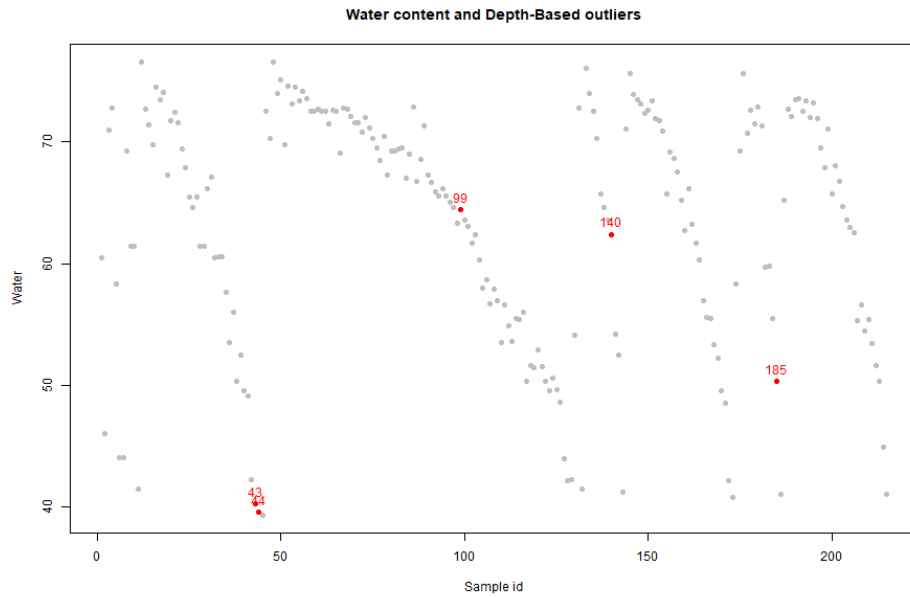
Samples 43 and 44 are especially notable because they have very high fat content, around 48%, and low water content, around 40%. Their spectra are therefore atypical in a way that is consistent with their chemical composition.

Samples 99 and 140 differ from the high-fat outliers. They do not have extremely high fat content, but they have relatively high water values, and sample 140 also has low protein content. Their detection suggests that the functional boxplot is not simply identifying high-fat samples, but can also detect spectra that deviate from the central pattern in other ways.

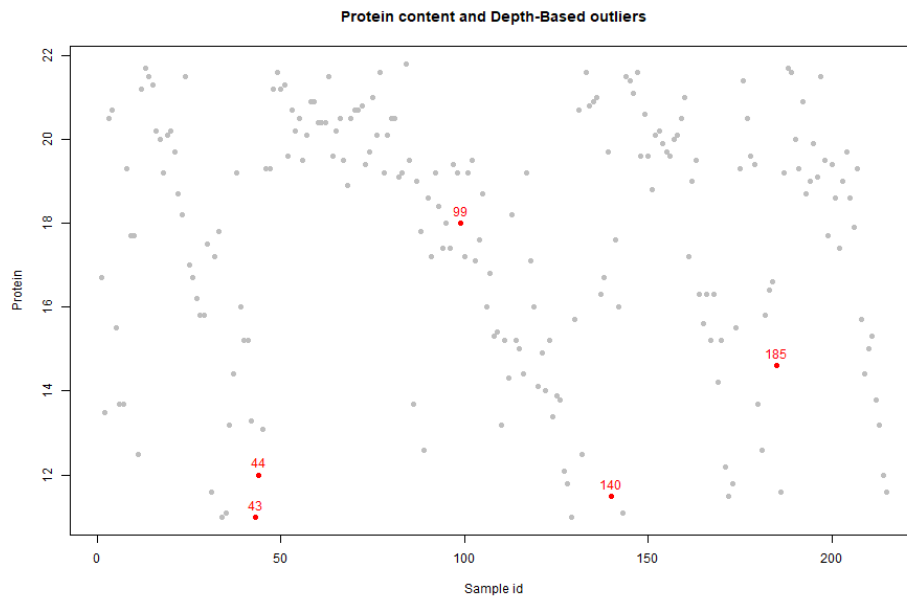
Sample 185 has intermediate chemical values compared with the other outliers, but still differs from the average non-outlying sample. This may indicate a spectral shape difference that is not fully captured by one chemical variable alone.



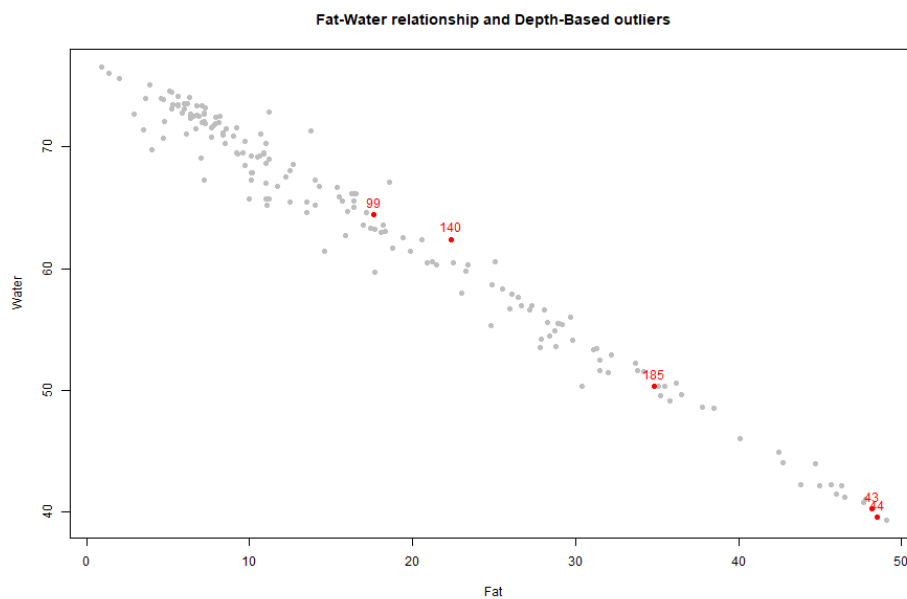
**Figure 8:** Fat content of all samples with depth-based outliers highlighted in red.



**Figure 9:** Water content of all samples with depth-based outliers highlighted in red.



**Figure 10:** Protein content of all samples with depth-based outliers highlighted in red.



**Figure 11:** Fat-water relationship with depth-based outliers highlighted in red.

Overall, the chemical interpretation supports the usefulness of functional depth methods. The outliers identified from the spectral curves are also unusual in terms of chemical composition.

## 7 Discussion and Limitations

The analysis shows that Tecator spectra are well suited for Functional Data Analysis. Each sample is naturally represented as a curve over wavelength, and the methods used in this report allow us to study the data at the level of the whole function rather than at isolated wavelength points.

The FPCA results show that the dominant mode of spectral variation is captured by the first principal component. The association between FPCA scores and chemical variables suggests that part of the functional variability is related to fat, water and protein content. At the same time, the interpretation of later principal components should be treated cautiously because they explain much smaller proportions of total variation.

Depth-based analysis provides a complementary perspective. While FPCA focuses on directions of variation, depth measures provide an ordering from central to outlying curves. The functional boxplot identified five outlying spectra, and these outliers were chemically meaningful.

An important advantage of the depth-based approach is interpretability. The deepest curve acts as a functional median, the central region summarizes the main functional behavior, and the outliers can be visualized directly as curves outside the functional boxplot fences.

This project also has several limitations. First, the functional representation depends on the choice of B-spline basis and the number of basis functions. A different number of basis functions could lead to slightly different smoothed curves and possibly different results.

Second, Modified Band Depth is only one possible depth measure. Other depth measures may emphasize different types of centrality or outlyingness. A more extensive study could compare Modified Band Depth with other depth methods.

Third, functional boxplot outlier detection depends on the fence rule. The use of a 1.5 inflation factor is analogous to classical boxplots, but it remains a methodological choice.

Fourth, the chemical interpretation is associative. The analysis shows that detected functional outliers have different average chemical composition, but it does not establish causal relationships.

Finally, the project is exploratory. Future work could extend the analysis by applying functional regression to predict fat, water or protein content from spectra, or by comparing depth-based outlier detection with model-based approaches.

## 8 Conclusion

This report applied Functional Data Analysis to the Tecator absorbance spectra. The dataset is suitable for FDA because each meat sample is represented by a curve over wavelength. The spectra were represented using a B-spline basis, then analyzed using FPCA and depth-based methods.

FPCA showed that the first principal component explains almost all of the functional variation among the spectra. The FPCA scores also show associations with chemical variables, especially fat and water, although later components should be interpreted cautiously because they explain much smaller amounts of variation.

Modified Band Depth and the functional boxplot provided a way to rank spectra by centrality and detect atypical curves. The functional boxplot identified five outlying spectra. These outliers were chemically meaningful: on average, they had substantially higher fat content and lower water and protein content than the non-outlying samples. Therefore, the depth-based analysis did not only detect visually unusual curves, but also highlighted samples with different chemical composition.

Overall, the project shows that FPCA and depth measures provide complementary tools for exploratory Functional Data Analysis. FPCA summarizes dominant modes of variation, while depth measures define centrality, functional median and outliers. Together, they provide a clear and interpretable framework for studying spectral functional data.

## References

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